

Date	Person	Log
Lent + Easter Term	YF	Made the De Novo IDP Generator Programme Modified the Protein Batch Analyzer - Made it pure CLI, included parallel processing, added new parameters Worked on NEXT Tag computational mutagenesis pipeline Made the Reverse Translation Programme
2026-07-01 to 2026-07-03	YF	Identifying Reagents and Filling Order Forms Discussed Ampicillin vs Carbenicillin - Decided on carbenicillin due to pricing and half-life Discussed liquid batch selection method - Half life of antibiotics => tetracycline may need to be constantly feed Discussed alternative selection methods for Tardigrade system - Using a fluorescence based assay instead of LDH Discussed tardigrade mechanism of action and truncation effects
2026-07-06 - 2026-07-10	YF	Made initial timeline draft for project - Considered total steps and rough timelines to create pseudo-gantt chart organization Discussed alternative selection methods for Tardigrade system - Searched for suitable fluorescence protein - Came up with survival assay instead Minor changes to selection method - Discussion of viability of batch selection due to extremely good variants rescuing others - Non-issue due to presence of proteases and rationale of average performance of pool still higher - Discussed suitable arabinose concentrations for plates - 0.2% for lac operon level expression - Max conc ~ 2% - from a paper - Decided to go for 0.2% initially and then increase it to increase selection pressure - Began to note volumes & concentrations for protocol Implemented melloid bin sorting into reverse translation programme - Sorts degenerate sequences into bins where they are similar to each other - aim: total sequence space across bins are lower - Instead, picked sequences with lowest loss and lowest

		encoding space Discussed viability of computational pipeline - Doubts on whether computational pipeline is reflective of reality => Too far along to change Wet Lab - Made chemically competent cells - Transformed chemically competent cells
2026-07-13 to 2026-07-17	YF	Investigated STOP codon appearance in RT programme - Fixed bug Investigated performance of degenerate sequence - Done through random sampling of degenerate sequence and filtering - Very low performance Investigated alternative methods of optimization - epPCR - James - Will make simulation latter - Shortening hotspot region - Kean Ann - Not significant change - Proportion Optimization: significant change - proceeded Investigated problem of hotspot optimization - Proportion Optimization considers all mutations as equally likely => Introduced MAD optimization on noise to identify hotspots Refined planned workflow for first 2 weeks - Determined to sequence at every step of pipeline - Decided against qrtPCR and other validation steps due to 'fluke' rationale - can't eliminate all & 'fluke' equally affects all variants - Identified need to transform into DH5a for plasmid concentration amplification (1st round Gibson max conc ~1.5ng/ul - require 10ng/ul for plasmid optimization) - Considered workflow from transformation to selection Wet Lab: - Practiced using nanodrop machine - Did plasmid extraction - Did gel electrophoresis of plasmid digests
2026-Lent to 2026-07-17	CW	Researched on 2 potential themes: directed evolution of G-quadruplex DNAzyme to catalyze C-H amination reaction, and directed evolution of intrinsically disordered peptide as solubility-enhancing tag for recombinant proteins that aggregate into inclusion bodies. Decided to do the second theme because finding a functional DNAzyme variant requires more time.

		Invented the SSB-derived IDP tag: inspired by the intrinsically disordered region close to C terminus of E coli Single Strand DNA Binding Protein, and literature said that without this region the Single Strand DNA Binding Protein aggregates more into inclusion bodies. Invented the beta-lactamase selection system: inspired by mutagenesis experiments on beta-lactamase. Mutations that lead to a more aggregation-prone hydrophobic surface in the folding intermediate stage might be rescued by solubility-enhancing IDP. When applying antibiotics as selection pressure, better IDP causes more soluble beta-lactamase, more antibiotic resistance, so the E coli survives after selection. Made preliminary plasmid construction plan for expressing the beta-lactamase-IDP fusion protein: Start from plasmid pACYCduet-1, use golden gate (restriction enzyme includes BglII, AatII, XmaI), assembled to insert beta lactamase–GGGSGGGGS linker–IDP tag under control of T7 promoter (which overexpress the protein to create aggregation tendency). Method did not go through due to lack of pACYCduet-1 plasmid. Helped revise plasmid construction plan for expressing the beta-lactamase-IDP fusion protein: decided to put fusion protein under control of arabinose-induced promoter. Still stronger than natural beta lactamase promoter, creates aggregation tendency. Better than T7 because it would not cause all fusion protein in inclusion body, allow differential solubility by different IDP variants, allow differential survival of E coli when antibiotic applies. Also decided to use Gibson assembly instead of golden gate because there are no good enzyme that cut exactly at the site to insert and optimally express fusion protein. Also decided to directly synthesize aggregation-prone beta lactamase gene instead of performing site-mutagenesis on normal beta lactamase because it would be time consuming and significant mispriming were shown during site-mutagenesis primer design.

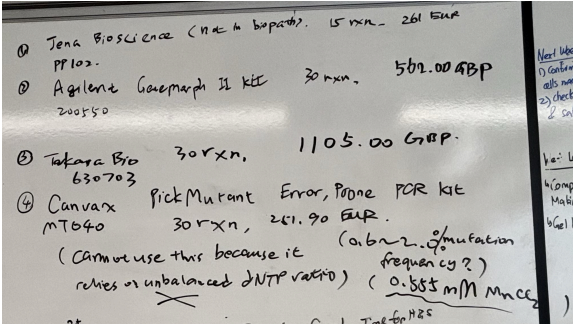
Date	Person	Log
2026-07-20	YF	Introduced 'Project Diary' for Task Keeping Discussed epPCR Buffer: Kit with Mutagenic Nucleotides vs Homebrew with dNTP - Reviewed expected variation between both - Identified that homebrew will have major unmutated version - Argued that its better to optimize for lack of original sequence than it is to optimize for mutation rate as no idea effect of mutation rate on performance Reviewed budget to see if feasible to buy kit - Realized the ds Degenerates are significantly expensive - Opted for ss synthesis at risk of library shortening during PCR - non-issue as lack Gibson flanks to assemble - Brief discussion on final target protein (aggregation prone protein, enzyme - LDH [solubility + function], aggregation prone protein with tag - rescue experiment) - Action Points (General): secure sequencing deal, identify running buffer, finalize protocols for other Discussed Tardigrade Protocol to Check Reagents - Identified we should have a control to compare performance of tag against regular cryoprotectants (on second thought, is there a need?) - Used to validate that the e coli can be rescued from desiccation - have to find suitable - Discussed if we should have an alternative IDP to show that improvement is strictly due to CAHS - not necessary Checked with Siol on Miscellaneous Tasks - Reagents: asking to get Taq kit - not making homebrew version due to precision and instead receiving expired kit - DPN1: Siol is checking for us - COSHH forms: to be sent blank and signed copies - Health & Safety Meeting: time to be confirmed - Competent Cells: DH5a, XL1 Blue, Checking if BL23 exist - Oligo ordering: two minor problems - Tardigrade motif 1 G block too short, degenerate double stranded Working on Infographics & Data Presentation for Computational Pipeline - Produced sketch infographic for computational pipeline

2026-07-20	Setareh	1. Searched for comparison of mutation rate between baseline PCR with taq polymerase and epPCR with differing conditions of magnesium, manganese chloride and biased dNTPs and the approximate increase in mutation rate is around 600-800 folds. 2. Researched about the prices of raw ingredients of MgCl₂ and MnCl₂ prices in comparison for the very small amount needed to make the epPCR solutions and had a discussion and concluded that ordering the kit is epPCR kit is eventually around the same price and more affordable as were as time saving and preventing errors. 3. How the concentration of manganese chloride specifically affects epPCR rate: Findings were mixed as some papers used both taq polymerase and pfuS polymerase In one such experiment it was found that around 0.75-0.8 mM of Mn²⁺ was optimal. It was also found that that concentration produced 60 mutations per kb Some other papers also noted an increasing mutation rate up to 60 per kb. We will probably try to optimise the concentration of Mn²⁺ and DMSO to increase mutagenesis rate without impacting the PCR product yield instead of ordering the kit according to our discussion.
2026-07-20	Kean Ann	Worked on tardigrade protocol: 1. Conditions needed for the experiment e.g. dry slowly so don't kill cells, need to express enough proteins before desiccation (pre-conditioning) 2. Verified species choice/method validity with Boothby et al 3. Discusses the need for alpha synuclein as a control → not necessary a. Very long (140 aa long so expensive) b. Don't know if it's necessary 4. Made sure that BL23 is a safe substitute for BL21 if BL21 is not available a. BL23 is better because lack the variant (recA has been mutated) that leads to recombination problems b. But might have a lower rate in protein expression/growth 5. Trehalose may be a viable control for the LDH assay a. Similar mechanisms to CAHS (Boothby et al, 2017), but the extent to which this is true is debated because trehalose is a disaccharide and CAHS is a large protein 6. Osmotic stress

		a. Read Bino et al and decided to use varying concentrations of sorbitol / mannitol / NaCl as a stressor
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2026-07-20	CW	1. Changed SSB, NEXT, De Novo, and Tardigrade tag degenerate library into ordering for single strand ultramer instead of duplex ultramer based on cost (278 GBP/seq to 132 GBP/seq to <90 GBP/seq). 2. Designed reverse primers for making ssDNA library into dsDNA ready for gibson assembly. 3. Decided to order only 7 nmol (3 OD minimal) of reverse primers for SSB, NEXT and De Novo Library in total, because dsDNA needed from each tag <0.5 pmol in gibson assembly, no need to convert 4 nmol of library per tag all into dsDNA 4. Decided to order 6.681 nmol (3OD minimal) of reverse primer for tardigrade motif 1 library. 5. Decided to increase "DOUBLE STRAND DNA non-degenerate Tardigrade fragment 1" from 114 base pair to 125 base pair for ordering as gblock gene fragment. Considered that "Primers to error-prone PCR fragment 1 (encoding motif 1)" will anneal correctly to amplify the 114 bp despite the lengthened 125 bp template. Checked that this is true by primer premier because lengthened area does not introduce new mispriming sites. 6. Also checked that the original template is able to be gibson assembled if the T5 exonuclease degrade 125-114=11 more nucleotides on 5' end, but this original template will be very low concentration compared to the mutated amplified 114 bp products. 7. Finalized and ordered with help from Dr.Sio, the following oligo: a. SINGLE STRAND DNA SSB Library b. SINGLE STRAND DNA NEXT Library c. SINGLE STRAND DNA De Novo Library d. SINGLE STRAND DNA DEGENERATE Tardigrade fragment 1 e. primers to make dsDNA from degenerate tardigrade fragment 1 (need HOT START PCR) f. DOUBLE STRAND DNA DEGENERATE Tardigrade fragment 2 g. DOUBLE STRAND DNA non-degenerate Tardigrade fragment 1 h. DOUBLE STRAND DNA non-degenerate Tardigrade fragment 2 i. Forward Primers to error-prone PCR fragment 1 (encoding motif 1) [error 1 fwd] j. Reverse Primer to error-prone PCR fragment 1 (encoding motif 1) [error 1 Rev] 8. Labelled the homologous region for gibson assembly in "Updated July 13th oligo ordering sheet" for newly ordered tags after July 16th. Sanity checked that all sequence theoretically will assemble correctly. 9. Researched the component of epPCR buffer (Tris-HCl+KCl), but realized it comes with the standard Taq polymerase. Therefore epPCR from Dr.Sio's side we already have:
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		a. Free standard taq polymerase b. DMSO to reduce mispriming c. MgCl₂-containing buffer d. Tris-HCl, KCl containing buffer e. Evenly mixed dNTP in the separately purchased box 10. We still need MnCl₂ solution, tentatively assigned as this form: https://www.sigmaaldrich.com/GB/en/product/sigma/m1787 10 x 1 mL MnCl₂ 57GBP? Any way to get less because this is way more excess? 11. Still need to finalise the protocol to maximize the mutation rate when using Taq. Higher Mn²⁺ concentration at 0.8mM? Higher concentration of Taq to overcome Mn²⁺ sensitivity and increase PCR yield? How to adjust a Mn²⁺ concentration based on Pfu/Taq mixture mentioned in Saterah's research paper? 12. What is the mutation rate we want, can we really achieve 60 nt/ 1kb template mutation without using the expensive nucleotide mutagenic analogs in kit? Check again the paper, and find the best condition that suits our needs. 13. Confirm NOT to use the expensive error-prone PCR kit, both from Agilent and from Takara bio. (230-1000 GBP/kit, 30 reaction)
2026-07-21	YF	Reviewing Lab Work Procedure for Plasmid Construction & Insertion Reorganizing Master Doc - Completed computational pipeline and data collation - Streamlined Master Doc Discussed Project Timeline - Realized it is not feasible to run three rounds of evolution + validation round => tentatively expect 2 rounds for now
2026-07-21	KA	1. Read a paper on using UV rays on tardigrades - realized that it is not viable because the 2011 paper did not go into details as to what proteins are used to protect the tardigrades against radiation - google suggests Damage Suppressor Proteins, which are different from CAHS 2. Tardigrade mispriming a. 3 piece-gibson assembly b. When a primer anneals to the top and bottom strand → conc of the long ones with an extra 11 bp is negligible c. Added the extra 11 base pairs to make it compatible with g blocks → highlighted in green in the ordering sheet 3. Checked tardigrade seq on ordering sheet

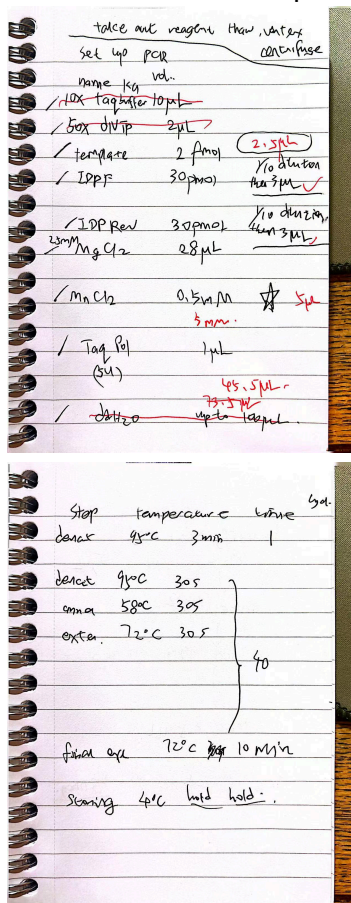
		4. Freeze thaw - Can't seem to find a paper that uses a whole-cell-freeze-thaw process - so if team really wants to proceed with freeze thaw, can try Kang's protocol for the LDH assay - Since we can't handle liquid nitrogen - use isopropyl alcohol + dry ice - first draft of method from google is listed in the workflow doc 5. Finalizing protocols for the tardigrade system
2026-07-21	CW	- Explained and checked that 125 bp fragment will be only amplified linearly while the majority of amplified product after epPCR will be correct 114 bp fragment that amplified exponentially. - Decided not to purchase any site-mutagenesis kit because they are too expensive. And the cheap kit has no special component compared to homebrew epPCR reaction system:  (cannot use this because it relies on unbalanced dNTP ratios) (0.62% mutation frequency?) (0.555 mM MnCl₂) - Decided to change Mn²⁺ concentration to reach the same level of mutation as the expensive kit. Researched the methodology and number of mutation per kb of gene appropriate for directed evolution of IDP tags from this article: https://doi.org/10.1038/s41589-025-02128-3 "The Mutagenesis GeneMorph II kit (Agilent) was used to introduce random mutations into the starting synIDP sequences. The 50-μl reaction mix contained the following components: 1 ng of DNA template, 250 ng μl⁻¹ of each forward and reverse primer, 5 μl of 10$\times$ Mutazyme II reaction buffer, 800 μM dNTP and 1 μl of Mutazyme II DNA polymerase. The PCR amplification was carried out in a thermocycler (Bio-Rad) under the following conditions: an initial denaturation step at 95 °C for 2 min, followed by 15 cycles of 95 °C for 30 s, annealing at 56 °C for 30 s and extension at 72 °C for 1 min per kb. A final extension step at 72 °C for 10 min was performed to ensure complete product formation. After the PCR program finished, 1 μl of unpurified PCR product was diluted 100-fold and used as DNA template for the next round of PCR. This completed one round of random mutagenesis. At least three additional rounds of

	error-prone PCR were performed before Gibson assembly.” High mutation frequencies (>9 mutations/kb) are obtained by using 0.1–100 ng of input target DNA, where the highest mutation rates can be achieved using the lowest recommended target amounts. Mutation rates up to 16 mutations per kb have been achieved using 0.01 ng of target DNA. From GeneMorph II Random Mutagenesis Kit Instruction Manual Catalog #200550. so (1+3) rounds * 12 nt mutation / kb/ round = 48 nt mutation / kb. Researched the level of mutation reached using Mn2+ instead of Agilent kit from these articles: https://patents.justia.com/patent/6803216 https://genome.cshlp.org/content/genome/2/1/28.full.pdf “In the presence of 7 mM MgCl2, 0.5 mM MnCl2, 1 mM dCTP and TTP, and 0.2 mM dGTP and dATP, Taq incorporates 4.9 to 6.6 mutations per kb per PCR” so 6 nt mutation / kb / round, need 8 round of PCR per IDP tag for epPCR to reach the same level (48 nt) as described in the paper. Finalized suggestion: Use standard Taq polymerase, adjust the buffer as described above, 8 rounds of (epPCR + then 100 fold dilution) for each IDP tag. To reach 48 nt mutation / kb / gene. Send the end-product for sequencing to see if really on average 48 nt has been mutated in product. If less than 48 nt, just conduct a few more rounds of epPCR using Taq by calculating how much mutation is introduced per round in actuality. Ask Saterah for the info: how many cycles per each PCR reaction does the 4.9 to 6.6 mutations per kb per PCR corresponds to? 4. Confirmed the ordering of TAE buffer as a universally good buffer for separating the correct PCR product from the incorrect ones. 5. Searched up and confirm the ordering of 4 items based on our budget. Ready to conduct epPCR once arrived. <table><tr><td>211726</td><td>Magnesium(II) chloride, 1M aqueous solution (50mL)</td><td>ThermoFisher</td><td>J63150-AD</td><td>42.95</td><td>1 set (50 mL, as ordered)</td></tr><tr><td>211726</td><td>FastDigest DpnI</td><td>Biotech stores</td><td>FD1703</td><td>16.77</td><td>1 set (50 units)</td></tr><tr><td>211726</td><td>Taq DNA polymerase, recombinant (50U/L)</td><td>ThermoFisher</td><td>EP0401</td><td>50.25</td><td>1 set (100 units)</td></tr><tr><td>211726</td><td>UltraPure 10X TAE Buffer</td><td>Biotech stores</td><td>15508042</td><td>11.10</td><td>1 set (1 L)</td></tr></table> 6. Arranged pour plate and start making competent cells for XL1-Blue, BL21 and DH5 alpha strains of E coli on	211726	Magnesium(II) chloride, 1M aqueous solution (50mL)	ThermoFisher	J63150-AD	42.95	1 set (50 mL, as ordered)	211726	FastDigest DpnI	Biotech stores	FD1703	16.77	1 set (50 units)	211726	Taq DNA polymerase, recombinant (50U/L)	ThermoFisher	EP0401	50.25	1 set (100 units)	211726	UltraPure 10X TAE Buffer	Biotech stores	15508042	11.10	1 set (1 L)
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		July 22nd at 9:30 AM.
2026-07-21	SS	Did research on epPCR and added some papers to the document called research about epPCR. Found varying papers with varying conditions of epPCR without using an epPCR kit. Based on the available data discussed and decided to use a manganese concentration of around 0.5 mM to optimise the mutation rate of epPCR without needing a special buffer for epPCR kit. Some papers had used higher concentration of Mn²⁺ however it was also shown that very high concentrations of Mn²⁺ can decrease yield of PCR product. Decided on the value because we are using Taq polymerase which has a higher sensitivity to Mn²⁺ than Pfu polymerase commonly used in some papers. Researched the level of mutation reached using Mn²⁺ instead of Agilent kit from these articles: https://patents.justia.com/patent/6803216 https://genome.cshlp.org/content/genome/2/1/28.full.pdf “In the presence of 7 mM MgCl₂, 0.5 mM MnCl₂, 1 mM dCTP and TTP, and 0.2 mM dGTP and dATP, Taq incorporates 4.9 to 6.6 mutations per kb per PCR” Therefore we are aiming for at least a minimum of 5 mutations per kb per PCR reaction under these conditions. Helped with comparing different Taq polymerase kits and discussed with Chunsu. Final decision was to purchase a Taq polymerase with two buffers of KCl and ammonium sulfate and a separate solution of magnesium chloride in order to be able to vary conditions and concentration if needed.
2026-07-22	YF	Wet Lab: - Poured Tetracycline Plates - Plated DH10B Refining Timeline for W1 & W2 - Determined which steps require gel electrophoresis - Determined order of steps for plasmid construction & transformation experiments Reviewed Viability of Liquid Batch Selection & Plate Selection - Key Assumptions that Need to be Validated - Different Concentration of Starting Population will lead to significant differences in final population given not saturated - Dead bacteria don't contribute to pellet size - Pellet formed is significant to be measured - The more I think the more problems come :(

22-07-2026	KA	Wet lab: 1. Poured Tetracycline Plates 2. Plated BL21 Tardigrade protocol 1. Proposed a new mech for desiccation a. Filter paper + dessicator + WT & variants in parallel etc., details in the relevant doc 2. Draft for verification step outlined 3. Reviewed steps that require gel electrophoresis
22-07-2026	SS	Wet lab practice = 1. Poured tetracycline plates 2. Plated XL1 cells
2026-07-23	YF	Wet Lab - Isolated 3 DH10B colonies to incubate - Made CaCl₂ solutions for making competent cells tomorrow Finalized Week 1 Project Timeline Working on Week 2 Project Timeline
2026-07-23	SS	Wet lab= - Isolated XL1 colonies into falcon tubes to incubate - Made autoclaved CaCl₂ and CaCl₂ with glycerol buffer solutions for use for making competent cells the next day
2026-07-23	KA	Wet lab 1. Isolated 3 BL21 colonies to incubate a. Grew into a murky substance in LB broth after incubation 2. Made CaCl₂ solutions for making competent cells tomorrow Worked on tardigrade protocol 1. Refine desiccation steps based on Siol's advice + what's available a. What's available: 7.0 cm whatman no.1 paper, hole puncher, a dessicator that can fit 7 petri dishes b. Do you even need sterilization??
2026-07-23	CW	Wet Lab: - Isolated 3 DH5a colonies to incubate - None grew => even Rachel's didn't grow => DH5a batch is bad
2026-07-24	KA	Wet lab 1. Made 120 aliquots of BL21 chemically competent

		2. Max OD: tube 3, 0.7 ish at 1:20 pm 3. Came up with a rough timeline using claudel based on the current tardigrade workflow
2026-07-24	SS	Wet lab = Followed the initial steps of the protocol for making chemically competent cells with XL1 XL1 culture did not increase in OD after a few minutes to hours Error was not mixing the incubated culture properly Not able to make competent cells
2026-07-24	YF	Made 80 aliquots (50 uL) of DH10B chemically competent cells
2026-07-24	CW	Started protocol to make chemically competent - Ignored -ve control failure => stopped during OD measurement phase

Date	Person	Log
2026-07-27	General	Measured pBAD conc: ~ 65 ng/uL Did pBAD linearization Began SSB epPCR - Discovered NEXT original tag is wrong - Finished PCR purification R1
2026-07-27	YF	Made 4 Gels: - 1 x 0.75% TAE Gel - pBAD linearization validation - 3 x 3% TAE Gels - epPCR validation
2026-07-27	James	- Created pitch deck under sponsorship - Organised meeting notes - General lab tasks, assisted with teammates in wet lab 1kb DNA ladders we ordered are gone somehow, will attempt to continue looking for it. Siol is giving us 100ul for now Will meet with IDT and imperial team in coming days
2026-07-27	CW	Started first round of epPCR.  <i>Handwritten notes:</i> take out reagents than mix, mix set up PCR name to vol. 10x Tagase 10 µL Box dNTP 2 µL template 2 fmol (2.5 µL) IDP 30 pmol No dNTP IDP Rev 30 pmol No dNTP 2mM MgCl2 2.8 µL MnCl2 0.5 mM 5 µL Tag Pol (5U) 1 µL dH2O 45.5 µL up to 100 µL Stop temperature time denat 95°C 3 min 1 denat 95°C 30 s anneal 58°C 30 s exten 72°C 30 s } 40 final ext 72°C 10 min cooling 4°C hold

liquidify next orig
 81.5 fmol / μ L 25 μ L
 ↓
 take 1 μ L
 ↓
 add 99 μ L dH₂O
 ↓
 take 2.5 μ L
 same procedure for SSB orig.

1 μ M → 0.5 $\times$ 10⁻³ M
 0.5 $\times$ 10⁻³ M
 1 μ L stock
 1999 μ L H₂O added
 2000 μ L
 add references to comment
 section for tracing back.

Eluted the dried SSB original DNA gblock ordered into 10ng/uL solution for future direct pipetting.

Further dilute 1 uL of SSB original DNA template by 1/100 to get 0.9 fmol / uL, 100 uL solution in a small PCR tube. Stored at freezer -20C, labelled 1/100 SSB original.

Dilute 1 uL of IDP forward & reverse primer by 1/10, to get 10 uM, 10 uL primers for pipetting 3 uL primer into the PCR system. Without dilution you need to pipette 0.3 uL from 100 uM solution, too inaccurate. Diluted primers stored at -20C in the freezer.

Found out wrong sequence in the original NEXT tag due to deletion of one base in the master document, have to re-order it but waiting to see if IDP is willing to sponsor us for free on that. UFS break down so at least that epPCR need to run with the degen seq's PCR in parallel. If IDT is willing to sponsor, will cancel the order before Friday and tell Dr. Sio.

		Guided the use of the protocol mastersheet and the process of pBAD linearization PCR. Final system and evidence of checklist in adding all reagents into the PCR reaction, as shown in here. Eluted the MnCl₂ for epPCR at 10 mM concentration, with 2000 uL volume in total for future epPCR use. Stored at 25C in the 5 mL large tube. Modified tomorrow's gel electrophoresis system to: make 2% agarose gel, every buffer used is TBE, run 70V till whenever the bromophenol blue band in loading buffer reached 2/3 of total gel distance.
2026-07-27	SS	1. Did purification of the epPCR generated products 2. Decided to leave Purification for pBAD linearisation the next day so we could check with gel electrophoresis before carrying out digestion
2026-07-27	Kean Ann	1. Did pBAD linearization 2. Found out conc of Pbad is roughly 65 ng/microL 3. Made slides for the tardigrade section for IDT sponsorship 4. Discussed some details of the tardigrade protocol with the team
2026-07-28	General	Ran Gels for epPCR & pBAD - pBAD resolved nicely => proceed with purification - epPCR has smear => troubleshooting required pBAD DPN1 digest & purification failed epPCR purification failed - Troubleshooting purification - transition to gel extraction (gel extraction damaging DNA - confirmation?) Learning gel extraction. Gels have been cut, will be digested and extracted tmr using kit
2026-07-28	KA	1. Spent the whole morning on Gibson Assembly protocol (confirming volumes, masses etc.) but did not end up carrying it out because the purification results are looking a bit depressing 2. Confirmed gel results for pBAD linearization were successful (very bright band at 4000 to 5000 bp, some faint bands at 8000 bp, likely due to long linearised plasmids entangling with each other) 3. Did loading exercise + the first part of gel extraction (cutting gel under UV) 4. Also made into liquid the solid powder for beta

		lactamase insert by adding 10 microL of TE into it, incubating at 50 celsius so that its conc is at 100 ng / microL, it is then stored in the freezer 5. Researched on details of the tardigrade dummy test + prepared workflow for tmr
2026-07-28	SS	1. Confirmed pBAD linearisation successful using gel electrophoresis and viewing 4000 bp bands against the 1 kb DNA ladder. 2. Nanodrop to find the concentration of DNA in the linearised pBAD product and confirmed an average of 390 ng/uL in all three samples. 3. Manufacture protocol for DpnI digest required around 0.2 ug of DNA in each 30 uL tube, calculated the dilution values required for carrying out the digestion. 4. Did 1/20 dilution of the original solution and pipette 10 ul of the diluted solution into the reaction tube to achieve a total of 0.2 ug. 5. Carried out purification on digested pBAD. 6. Results of purification are not good. Purification of epPCR products produced only around 30 ng/uL of purified solution and even lower for digested pBAD of around 9 ng/uL and both had a low 260/230 nm value indicating impurity.
2026-07-28	YF	Loaded gels for epPCR & pBAD - Both gels manage to run (DNA ladders resolved) - pBAD +ve result => proceeded with DPN1 digest - epPCR -ve result => troubleshooting Learnt cutting gels => will gel extract tmr
2026-07-28	James	Poured gel for gel extraction and gel run tmr @0.8% TAE. SYBR stains are faint, may need to be added in greater concentration/cooler temps. Did not carry out gibson assembly and transformation. Competent DH10 strains and tet plates are available from previous week's work
2026-07-28	CW	Research potential reasons and designed troubleshooting experiment for epPCR. Measured the epPCR unpurified product concentration seems very high (although not sure how much is due to primers and dNTP). Therefore conclude our purification has a problem. Re-run purification and tested whether the buffer washed off the DNA binding column contains any DNA. Result shows that likely DNA was retained inside the column after elution. Therefore would require higher temperature and longer time to let DNA dissolve in the elution buffer. Suggested to change the brand of DNA binding column for

		both gel extraction and pcr clean up kit. Particularly gel extraction kit.
2026-07-29	General	pBAD linearization + digest ran on gel electrophoresis - Clear band seen - Gel extraction failed - Speculate might be low conc of starting gel => ran pBAD linearization at higher cycle no. to increase conc pBAD linearization re-ran to troubleshoot gel extraction epPCR ran on 2% Lithium Borate for post-staining method - Smear still seen Gel extraction consistent fails across different samples and different people - Might be kit error Began preparation for tardigrade dry run
2026-07-29	KA	1. Finalised some details of tardigrade protocol 2. Cultured some cells from the ones made chemically competent days ago 3. Did some OD measurements but they were still less than 0.1 after 3 hours (at 2pm ish), the control seems to be contaminated 4. Made 4 more tubes (15ml and 5ml and 2 controls) for overnight incubation at 2.30pm, will check results tmr 5. Poured plates with no antibiotics (with help from Chun Ting) 6. Did Pbad linearization again for 35 cycles instead of 30 to increase conc of products, seems like the problems after purification stem from too low of a conc, results from gel electrophoresis seem to show that the brightness of the bands is too faint, will check gel results after DPN1 digestion 7. Did not carry out Gibson Assembly because the nanodrop results from gel extraction are not looking good.
2026-07-29	SS	1. Carried out DNA purification once more on epPCR and digested pBAD with more thorough mixing of binding solution and DNA in a separate tube however the results did not change or improve. 2. Think there might be a problem with the purification kit or there might be a problem with the low concentration of initial DNA we are using. 3. Checked and worked on the protocol for plasmid extraction and the protocol for the second round of pBAD with insert linearisation. 4. Think that the IDP insert tag is in solid form as provided by the manufacturer is only OD and yield in nmol, and the Merck website states that solid form oligos are ready to be suspended in either buffer or ddH2O. 5. Planned the draft of protocol for the linearisation, using first the oligo to create stock 100 uM solutions and use

		appropriate amount to get a final concentration of 0.5 uM in the 50 uL PCR tube. 6. However won't be able to carry out plasmid extraction or pBAD linearisation tomorrow because gel extraction or DNA purification did not work and therefore we didn't do gibson assembly.
2026-07-29	YF	Made 4 x 0.75% TAE Gels Made 1L 40x Lithium Borate Solution & 2L 1x Lithium Borate Solution Made 3 2% Lithium Borate Gels Ran Gel Extraction on Sample, failed (massive contamination) Loaded gels for epPCR and pBAD
2026-07-29	James	Finalised support from IDT. discounts available on future orders Completed gel extraction for both test sample provided by Siol, as well as DpN1 digests of pBAD linear products. - Low DNA conc (5~20ng/ul) - Low 260/230 (~0) - Varying 260/280 - Strong peak at 230 instead of 260, indicating contamination Ruled out operator error by having yf perform the same digest supervised by advisors. Will increase concentration of DNA inside the gel by running 35 cycle PCR instead of 30. Will redo DpN1 digestion of pcr products tomorrow, and gel extract again. If this still fails, then likely we have to get a different gel extraction kit. Check existing inventory, previous iGEM teams have some leftover New issue: gibson assembly instructions requires 200ng/ul DNA. with current methods, DpN1 digestion cannot produce such high concentrations. Tomn to (hopefully) come up with a revised protocol for both steps by tomorrow.
2026-07-29	CW	Suspect that epPCR product smear was not due to actual variation in DNA size, but due to SYBR binding and increasing resistance. Performed post-staining by placing the gel in 7cm x 7 cm box and add SYBR and swirl in incubator. Post-staining still smear, indicates the DNA size is probably genuinely different to cause the smear. Have been writing, revising and organizing the protocol mastersheet to adjust for temporary changes to standard protocols. E.g. use more amount of PCR product relative to Dpnl as long as the methylated DNA template has very low amount and can be fully digested by Dpnl.
2026-07-30	General	Reran pBAD linearization + digest product at higher conc - Clear band seen in gel electrophoresis - Amy did gel extraction => failed (suspect kit error)

		- Rebuy new kit (not compatible with borate systems => transition back to 3% TAE system) PCR product of SSB failed - Suspect either Taq failure, annealing temp failure or template/primer failure - Running troubleshoot experiment for Taq and annealing temp Running tardigrade desiccation dry run today
2026-07-30	YF	Made 3 x 0.75% Lithium Borate Gels Loaded gels for epPCR and pBAD Excised pBAD bands
2026-07-30	SS	Used the gel extraction kit to practise gel extraction. Nanodrop results showed a concentration of 10 ng / ul and a 260/280 of 2.24 and a 260/230 of 0.02 indicating high impurity. We think the gel extraction kit may not be working or there is a problem with the low concentration that we are using Helped set up epPCR program on machine Still working on the protocol for the second round of pBAD linearisation The phusion mastermix protocol states to use 72 degrees celsius for combined annealing and extension using primers more than 30 nucleotides and says that if amplification at 72 degrees does not give good results you could try a temperature gradient using different blocks and compare the results on agarose gel. Carried out purification on digested pBAD with DPN1 using modified protocol . The DNA concentration was still too low after purification at around 10bng per uL and the 260/230 measurement was too high at 3.33 and around 10 .
2026-07-30	KA	Ran the full desiccation protocol: https://docs.google.com/document/d/1ZRMLpL6hlfsiGlb1T7Hd6lfUiSlyM7y2_fb8p5UCoc/edit?tab=t.0 Refined some details of protocol based on today's mishaps and mistakes
2026-07-30	James	Purification failed. Will look into procuring new kits DpN1 digest: - pBAD linearisation products show ~390 ng/ul, 260/280 = 1.87, 260/230 = 2.33 - Generally promising results. Digestion of pBAD linearisation should give us a 1:3 dilution - After digestion: DNA = 139 ng/ul, 260/280 = 1.56, 260/230 = 0.93. Good results with contamination (from dpn1 kit) - After gel extraction: ~0ng/ul

		- This shows that digestion works, it is the gel extraction step that is having problems. Amy verified by carrying out the extraction herself to show the issue lies with the kit and not the operator Helped out with the dessication protocol. We decided to dilute by x1000 and x10000 based on calculations, but no colonies grew on the undiluted plates. Dilution was inaccurate - tried pipetting 495 out of 500 ul using p1000.
2026-07-30	CW	Using only conventional PCR condition, SSB original sequence was used as template again and primers are still IDP F and IDP R. Gel electrophoresis result show faint smear and no correct product size band. This suggests that not only mutagenic condition was contributing to the smearing. Could be due to: Taq polymerase expired and stop functioning. DNA template degraded. Primers were degraded or not present at sufficient concentration. Annealing temperature was incorrect→most likely culprit.
2026-07-31	General	epPCR troubleshooting results: - Phusion had 2 products at 57.7 celsius => primers & annealing temp problem - Misprime product disappeared at upper boundary of testing temps => optimized temp at 61 celsius
2026-07-31	YF	Made 4 x 0.75% Li3BO3 gels Loaded PCR variable temp & taq troubleshooting experiments (x2) - Found Phusion had 2 mispriming events at 57.7 celsius (annealing) - Found for Taq, at higher temps, less mispriming Loaded for Higher variable temps - No great difference at higher temps => now need to optimize Mn & Mg
2026-07-31	SS	Poured LB-agar plates
2026-07-31	CW	Tested range of annealing temperatures for SSB original PCR. At lower annealing temperature e.g. 57.7 C, 2 product bands are observable. It was very confusing that the lower size product seems to have correct amplicon size, but the higher size product seems even longer than the original template. Anyway the mispriming product seems to vanish as annealing temperature increases. However technical issue seems to cause the PCR product annealed at ~60 C to smear more. Therefore our observed optimal annealing temperature was at ~61 C.

Date	Person	Log
2026-08-03	General	Conducted several rounds of Gel Extraction - Varying elution conditions gave significant results (10 mins at 65 celsius) Ran Gel Electrophoresis on Digested vs Undigested product to see if it interferes with gel extraction - Both contamination, not digestion process that causes error - Suspect PCR mix might be the error Made tetracycline & L-arabinose stock solutions Ran Mn & Mg optimization experiments - Gel came up blurry, rerun gel tmr - Have preliminary parameters based on gel Transformed BL21 with pBAD and incubating overnight with tetracycline
2026-08-03	YF	Conducted Gel Extraction on pBAD sample (excised YF 30/7) - Conducted with original protocol, 10 min elution incubation and 10 min at 65 celsius elution incubation - Last condition seemingly had great results; no triplicate, upon repeat 1 hr later, failed - Subsequent attempts cannot replicate success Loaded Undigested/Digested Troubleshoot experiment Loaded Mn/Mg optimization Troubleshoot experiment Made 4 x 2% Li3BO3 gels pre-stain 35 mL - Gave bad results - have to test again Made 3 x 3% TAE gels post stain 35 mL - Solidified quite fast - may not be viable
2026-08-03	KA	Came up with a protocol for 10 mg/ml tet + 20% l-arabinose stock solutions Transformed BL21 with pBAD + overnight incubation with tetracycline Refined some details on protocol Details in: https://docs.google.com/document/d/1ZRMLpL6hIfsiGib1T7Hd6IfUiSlyM7y2_fb8p5UCoc/edit?tab=t.0 Cut some gels
2026-08-03	SS	Conducted pBAD digestion and made six new samples. Carried out DNA purification using the kit again with changed protocol from incubation at room temperature for one minute to incubation at 50 degrees for 10 minutes for epPCR product purification and incubation at 65 degrees for 10 minutes for

		digested Dpn1 products. Results more promising than standard protocol. Achieved higher concentration of around 20 ng/uL in purified samples. If purified Dpn1 digested product is still consistent tomorrow we could continue with Gibson assembly or if not we could repeat experiment with new protocol.
2026-08-03	CW	After determining the optimal annealing temperature at ~61C, start testing the uppermost tolerable mutagenic stress condition. Tested gradual increase of Mg²⁺ from 2.5 mM to 7 mM. More Mg²⁺ neutralizes the negative charge repulsion between mismatched (mutated) base pairs. Also tested gradual increase of Mn²⁺ from 0.0mM to 0.5 mM. More Mn²⁺ increases the mistake that Taq polymerase catalyses. Threshold for tolerating Mg²⁺ is 4mM, and threshold for tolerating Mn²⁺ is 0.15 mM.
2026-08-04	General	Mn/Mg Optimization Experiment samples reloaded and gel resolved fine => optimized parameters for Mn/Mg is confirmed epPCR R1: 1 misprimed => gel excised for gel extraction and sample from tube was PCR purified for R2 epPCR R2: 5 used Taq before ran out, 2 used expired Taq - Resolve tmr pBAD digest & purification conducted, products used for Gibson and transformation which was plated for monoclonal selection tmr BL21 pBAD plated for monoclonal selection tmr Tardigrade dry run cells have been prepared and to be used tmr
2026-08-04	YF	Loaded: - Mn/Mg Optimization Experiment: clear bands - prob previous gel ran off - epPCR R1: 1 misprimed - gel excised for gel extraction Started freezer reorganization by reagent and products Started reorganizing pipettes (sterile, sizes, etc) for planning items to autoclave Conducted epPCR R2 using PCR purified sample from tube 1 from epPCR R1 Tried to sterilize plastic vessel for hydration control experiment for tardigrade
2026-08-04	SS	- Made six new Dpn1 digested samples from linearised pBAD and labeled as batch 2/ b2

		- Carried out two PCR purifications on two different Dpn1 digest samples with 10 minutes of incubation at 65 degrees at the elution step - Carried out one PCR purification on epPCR product - Very consistent results for concentration of around 21.8-25.9 ng/uL for epPCR as well as good 260/280 value close to 1.80 and 260/230 value close to 1.2 - Broader range for Dpn1 , but the best sample had an average of 19.5 ng/uL when discounting the anomaly and was the sample used for gibson assembly.
2026-08-04	KA	Added 1 ml of overnight culture to 50ml fresh lb broth + plated the negative control to check for cell growth + effectiveness of transformation Incubated to od = 0.498 and added l arabinose The cells glowed under UV light so l arabinose works Centrifuged and prepared the cell pellet for tmr Prepared 700ml of saline water Conducted gibson assembly 1. Used number of moles of pBAD = $\frac{1}{2}$ number of moles of b-lac and settled on 0.81 microL of b lac + 9.19 microL of pBAD
2026-08-04	James	Gibson assembly was conducted. Products were transformed into DH10b. Gibson results were promising, ~1300ng/ul on nanodrop with minor contamination. Transformation appears to have succeeded - liquid culture of transformed cells is cloudy while control is not. However plates had no colonies, will replate on 5/8
2026-08-04	CW	Using the new, less smearing gel electrophoresis result, we confirmed the tolerance threshold for Mg ²⁺ and Mn ²⁺ . Following this, epPCR round 1 on SSB original template, gel extracted, concentration measured, and epPCR round 2 was carried out.
2026-08-05	General	Investigated yields & purity of PCR vs Gel purified paper Investigated effects of purification on sample Conducted dry run of tardigrade desiccation run with saline
2026-08-05	YF	Loaded epPCR R1 products (10 uL) - Added template instead of PCR purified product - failed - 9 & 10 no product => pipetting error - Expired Taq works - All product misprimed Loaded purified epPCR Mg/Mn optimization product - Removed high bp length - Very low conc of bands

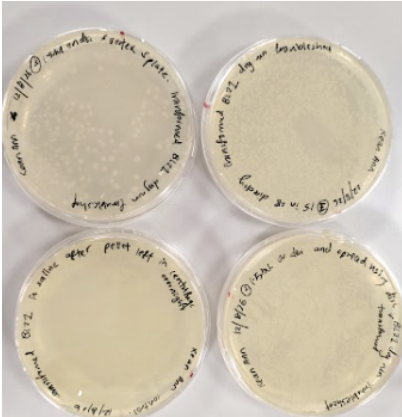
		Made 3 x 35 mL 2% Li3BO3 prestain gels Made 2L x1 Li3BO3 Gel Extracted epPCR R1 product - Analysis shows gel extracted product similar yield but less pure than PCR purified product Prepared spare tardigrade dessication dry run samples
2026-08-05	SS	- Made ten 50 mL LB lennox broth in flasks to be autoclaved - Ran 7 purifications on epPCR products from troubleshoot on 31/08
2026-08-05	James	Replated DH10b transformed with pBAD with b lactamase insert, with 10 and 100 fold dilutions Mispriming products from epPCR were sent for sequencing to confirm identities
2026-08-05	KA	Conducted desiccation dry run according to the protocol above + plated 1:1, 1:10 and 1:100 dilutions Result: only the 3rd dish has 15 colonies, with only maybe a couple colonies from the diluted sample from the 3rd replicate Plated 9 plates in total for the diluted samples + 3 plates for water controls; water controls have no colonies
2026-08-05	CW	epPCR still cannot persist and the amplicon brightness decreases drastically. Smearing occurred and the amplicon does not seem at the correct size. Therefore proposed to systematically re-examine the effectiveness of different methods of purifying the PCR products.
2026-08-06	General	Investigated PCR vs Gel Extracted Product viability for epPCR - Gel Extracted Product amplified but produces wrong fragment - PCR purified product failed to amplify => suspected insufficient volume due to high concentration of template (dilute next time) Troubleshooting Tardigrade T0 results - No colonies for diluted sample - Either fault in elution volume, vortex time or sample Received Sequencing Data to Analyse
2026-08-06	YF	Ran epPCR on SSB Gel Extracted and PCR product Made 3 x 35 mL 2% Li3BO3 Prestain Gels Loaded Gel & PCR purified products

		Made Fastq sequencing data processor
2026-08-06	KA	Troubleshoot tardigrade T0 results Changed elution volume and vortex time Worked on the actual protocol for the selection system on desiccation, osmotic stress, freeze thaw and heat shock Result (checked on the 10/8: no colony)
2026-08-06	CW	The SSB epPCR sample that showed multiple bands was gel extracted, calibrated to 2 ng/uL and sent off to plasmidsaurus for premium PCR library sequencing. The sample should have representative by-products of all types that emerges in our previous failed gel electrophoresis results. Therefore, the sequencing data should inform us all identities of by-product. It was discovered that the ~224 bp mispriming product is the major smear after the correct amplicon band. Based on these sequence, it is likely that primer first annealed correctly, then stall in the GC-rich repeats directly downstream of sense primer binding site, then detaches from the template and annealed to some unknown template and added a foreign sequence of ~24 bp, then the primer re-primers on the correct template and synthesized the entire strand. Hence, the major cause of smearing seems to originate from the GC-rich repeats within a specific region of the template. Therefore, GC enhancer (1M betaine) that equalizes the energy release during AT and GC base pairing was proposed to add to future epPCR reaction. DMSO that disrupts hydrogen bonds in misprimed situations was also proposed to add to future epPCR reaction.
2026-08-07	General	Power Outage => No Wet Lab work done General discussions on tardigrade protocol, mispriming products, timeline for W6 and cloning protocol

Date	Person	Log
10/8/26	General	pBAD plasmid extraction attempted - Low concentration obtained => reincubate overnight pBAD + b-lactamase plasmid strain incubated to be extracted tomorrow epPCR mispriming optimization experiments done: - Varying DMSO, betaine, extension time - Sample used: epPCR R1 => effectively 2 rounds of mutation - Optimal condition found to do R3 tmr epPCR mispriming analysis programme in progress
10/8/26	James	Transformed DH10b were successfully plated. Incubating overnight in liquid culture to perform plasmid extraction tomorrow Plasmid extraction of pBAD from BL21 - Low yield observed ~14 ng/ul - 260/280 = 1.75, 260/230 = 3.75 - Promising results, but may increase elution time/temperature as well as amount of cells used - Will redo tmr with more cells and prepare glycerol stock for future use Online meeting with twist bioscience account manager - Different system from IDT/Merck - Specify variance at each position and have twist build a library instead of providing a degenerate sequence - Much more expensive than expected, can look into using library design tools to get rough quotes - Better system more suited for what we are doing. It would have automatically filtered out stop codons, something we had to fix earlier on
10/8/26	YF	Nanodrop epPCR R1 alpha sample for epPCR misprime troubleshoot experiment Loaded gels of sample 1, 2, 3, 5 from epPCR Mn/Mg troubleshoot experiment Made 300 uL 100 bp ladder Excised: - Mn/Mg product - epPCR misprime troubleshoot product (1, 2, 6, 7, 8, 9, 10) - 6(d) was dropped on the floor - last resort backup

		Gel Purified Mn/Mg product & loaded to validate extraction - Tho we used borate system, extraction successful, will proceed with borate system for yield independent processes, will use TAE for yield dependent processes - Sample 3 misloaded => doesn't appear in results + more dilute than others Made 2L x1 Li3BO3 solution Made 2x 4x 35 mL 2% Li3BO3 prestain gels 35mL thin combs => max fill vol 20 uL 35mL thick combs => max fill vol 30 uL
10/8/26	SS	Made ten tubes of epPCR using Tomn's protocol and program Made DMSO stock solution and ran epPCR on 5 DMSO tubes and five non DMSO tubes using the programmes 50-55 for non DMSO and 36-40 for DMSO. For the first batch of epPCR in the morning, the extension time for DMSO samples were 15 s and the extension time for non DMSO samples were 10 seconds. The results showed that DMSO optimised the product and gave product size about 170 bp
10-8-26	KA	Worked on epPCR Mispriming Analysis Tool → .html file in drive - Overall not perfect but can identify key problems? Will need to stress test + cross-check with Tomn on future samples Meeting with TWIST - Diff system, more expensive, can use their tool: https://ecommerce.twistdna.com/app to get rough quotes
2026-08-10	CW	Reused epPCR round 1 template, tested different concentration of DMSO, Betaine (GC enhancer), and the combination of both. Betaine (GC enhancer) was borrowed from the Phusion Plus kit instead of a natural component in Taq kit. Also, we only assume the GC enhancer mainly contains betaine because the ingredient was not listed. The optimal condition was only use GC enhancer and at 1X concentration. Also, because the mispriming product is actually larger than the correct amplicon, tried to shorten the extension time such that the replication of mispriming product is incomplete while the correct product is complete. Tested at 30s, 15s, 10s, 7s, 5s. Turns out the best combination is extend at 10s and before that anneal at 61.7 C. This gives the brightest correct product band, and the mispriming product band almost diminishes. Smearing level is similar to the 200 bp band of DNA ladder loaded.

11/8/26	General	Plasmid extraction conducted - 1 sample ~ 200 ng/uL, subsequent attempts ~ 60 ng/uL - Suspected error in extraction => redo with fresh overnight cultures tmr Dpt of Pathology Sequencing Services - Require elution in water & specified primer/generic primers => will attempt elution in water tmr 2nd linearization failed (smear produced + no amplification) => suspect due to inaccurate conc used => redone today with correct concentration and will be verified tmr R3 epPCR conducted in 2 conditions (with & w/o DMSO) using gel extracted product from epPCR misprime troubleshoot sample 2 & 6 - Failed (smear produced + no amplification) - Unknown cause, has been replicated by a different person (Tomn) => to be verified tmr
11/8/26	YF	Conducted gel extraction of epPCR misprime troubleshoot product samples Nanodrop samples for epPCR R3 (samples 2 & 6) Made 3 x 35 mL 0.75% TAE prestain gels Made 2L x 1L Li3BO3 buffer Loaded pBAD + beta-lac linear product (failed), R3 epPCR product (failed) & Gel Extraction product (succeeded) epPCR simulation can only proceed after attaining R5 - 7 sequencing data to model mutation rate & bias (hold)
11/8/26	SS	Did plasmid extraction on both BL21 and DH10B samples . There may have been a problem with my Dh10B sample during extraction protocol as the final volume looked significantly more than 75 uL which was the original volume of the elution buffer. Inoculated 5 tubes of DH10B from single colonies on Dh10B pBAD + B lactam 1:1000 bacteria plates with tetracycline plus a control tube
11/8/26	KA	EPPCR troubleshoot (with and without DMSO) Result: had smear, suspected pipetting errors but Tomn redid it and the result was the same so suggests problem stems from something inherent to the PCR itself and not pipetting errors Rearranged Gel results based on dates + categories Plated pellet from the desiccation dry run suspended in saline → result the next day: a lawn

		Plated 1:1000 diluted saturated BL21 culture for future inoculation → a lawn so had to be replated
12/8/26	SS	The control from the inoculated tubes yesterday failed, likely due to accidentally picking up a colony from the edge of the plate when cooling the metal rod. Did plasmid extraction on samples from two tubes. The results were comparable with concentrations around 20-30 ng/uL, however this was much lower than our sample from yesterday. The main aim of the experiment was to see if elution with water or TE buffer produces significant changes in the concentration of the final product as we need to
12/8/26	James	2nd linearisation PCR worked with adjusted primer concentrations, gel produced bright bands. Sequencing of pBAD + bla ^r plasmid templates revealed that the correct insertion had taken place. 2nd gibson assembly can proceed provided we can produce enough pcr products.
12/8/26	KA	Plated 4 plates: 1. Suspended pellet undiluted in saline (got a lawn) 2. 1.5 microL on disc, no vortex, dragged around plate (got fewer colonies but fatter colonies) 3. 1.5 microL of pellet in saline, vortexed for 30s in 0.5 ml LB shaken b4 plating with 100 microL of that solution (got a lawn but not as thick as 4.) 4. 1.5 microL directly into LB with 100 microL on plate (got a lawn) 5. Plated 1:10000 transformed BL21 → got countable colonies  Learnt how to use 96 well plate reader + prepared a protocol for modelling the growth curve
12/8/26	YF	Made 4 x 35 mL 2% Li3BO3 prestain gels and 1 x 50 mL 0.75% TAE prestain gel Loaded - pBAD + beta-lac linearized product - failed (smear) - V2 (corrected primer conc) 2nd linearization - succeeded - excised - epPCR R1 + R3 - failed (smear)

		- epPCR R3 (DMSO + extension time) - failed
13/8/26	SS	Plasmid extraction on BL21 and DH10B strands. Nanodrop gave consistent results of around 30 ng/uL in concentration for both. Elution was carried out twice to maximise concentration, still not sure why concentration for DH10B specifically is quite low. Made 2 litres of 1X TAE buffer
13/8/26	KA	1. Carried out the dry run for growth curve, but wasn't successful because the controls were contaminated + power outage btw 1 to 2 pm disrupted things 2. Incubated the "controls" tubes but were cloudy the next day 3. Rearranged + reorganised the fridge and threw away some tubes from weeks ago
13/8/26	YF	Made 1 x 100 mL 2% Li3BO3 prestain gels and 1 x 50 mL 0.75% TAE prestain gel Loaded - 3rd 2nd linearization product - success - 2nd linearization DPN1 digest samples - extracted - epPCR troubleshoot (primer conc) - succeeded
14/8/26	General	Received sequencing data and found that plasmid sent was NOT pBAD but beta lac inserted pBAD
14/8/26	CW	Tested if lowering primer concentration would decrease smearing in epPCR. Because one negative control tube that does not have any template added seems to produce a smearing band <300 bp. Wonder if high primer concentration lead to primer-primer dimerisation or sth and produced by-product. Given that smearing seems inevitable in epPCR, recommended to do gel extraction to exclude by-product for all subsequent rounds of epPCR. Was not able to carry out some wet-lab procedures due to UV-light injury on skin.
14/8/26	SS	Made two solutions each 500 ml containing LB Lennox powder and agar or LB lennox powder and Miller. Protocol is to add 5 ml of L arabinose to make a 0.2 percent solution and 250 uL of tetracycline to 250 uL of solution . Poured 20 Miller agar gel plates , 10 containing L arabinose only labeled black and ten containing both l arabinose and tetracycline labelled black and red /
14/8/26	KA	1. Carried out growth curve dry run using 96 well plate reader to model bl21's growth in diff media but results

		are pointing towards dh10's typical behavior instead 2. Plated the cultured cells used in this growth curve dry run on plates containing l arab + tet → if glows: BL21; if doesn't: dh10 beta → awaiting results on Monday: results: it is DH10 beta, no glow under UV 3. Plotted a graph on python with error bars etc myself but cross checked with claude to ensure code's accuracy Figure 1 4.
14/8/26	YF	Made 1 x 100 mL 2% Li3BO3 prestain gels and 1 x 50 mL 0.75% TAE prestain gel Extracted 13 gels - epPCR - DPN1 digest Loaded - epPCR primer optimization gel extracted sample - epPCR primer optimization w/o G/C enhancer - DPN1 digested 2nd linear + water control

Date	Person	Log
17/8/26	General	Out of 3 'BL21' colonies selected and cultured from 3 of Kean Ann's plates, only 1 glowed under arabinose conditions. This means that the other two probably are DH10b colonies which contain the plasmid carrying the b lactamase gene instead of gfp. Linearisation PCR of the two stock solutions of pBAD and pBAD+b lac plasmids reveal that the stock plasmids are correct. Plated cells transformed on friday onto arabinose plates to confirm that they absorbed the right plasmids. Current plans for epPCR: - Can only do 1 round of epPCR - insufficient time to epPCR next sample - For future discussion
17-8-26	KA	- Conducted EPPCR troubleshoot
17-8-26	James	Transformed: - BL21 with pBAD+gfp - DH10B with pBAD+gfp - DH10B with pBAD+b lactamase Currently in liquid culture and in plates, will observe results on monday
17-8-26	YF	Spread 4 cultures on Tet + Ara Miller Plates (SS - 14/8) - BL21 + pBAD - DH10B + pBAD/beta-lac - DH10B + pBAD - Control Made 1 x 100 mL 2% Li3BO3 prestain gels, 1 x 100 mL + 2 x 35 mL 0.75% TAE prestain gel Loaded: - Linearization plasmid troubleshoot: plasmid tubes are correctly labelled - epPCR replicate - epPCR successfully replicated
17-8-26	CW	epPCR round 2 seems to work now, so the optimized condition was effective to some degree.
18-8-26	General	Had meeting with IDT & NEB - IDT: discounts on orders + credits for past orders - NEB: offered to give stuff depending on what we need/ask for Transformed cells for actual experiment obtained (Arabinose plates) - Correct plasmids have been inserted

		ssDNA to dsDNA conversion of all degenerate tags done Gibson assembly of SSB degenerate done, will confirm success tmr epPCR R3 conducted + troubleshoot unpurified vs purified PCR product performance
18-8-26	YF	Made 2 x 1 x 100 mL 2% Li3BO3 prestain gels (2g, 100 mL, 5 uL SYBER) Made 1 x 100 mL 0.75% TAE prestain gels (0.75g, 100 mL, 5 uL SYBER) Loaded - epPCR R2 sample 4,5,6 for gel extraction - epPCR replicate - Plasmid linearization repeat Transformed DH10b with Gibson product + control Did serial dilution and plated
18-8-26	KA	Conducted ssDNA → dsDNA of all degenerate sequences (20 microL in total for each tube) Conducted Gibson Assembly of SSB degen sequence → nanodrop result: 1100+ ng/microL; 20 microL of product, 20 microL of control that has ampicillin resistance
18-8-26	CW	Revised then further designed all protocols up to constructing the plasmid expressing the beta-lactamase-gs linker-idp fusion protein. Helped with ssDNA to dsDNA conversion of degenerate tags. The primers are the same as epPCR (IDP F, IDP R), and the amplicon length is also the same. To synthesize the complementary strand as high-fidelity as possible, less smear is required. Some optimized parameters were adapted from epPCR troubleshooting. Also helped with concentration calculation, molar ratio determination in Gibson assembly.
19-8-26	General	Transformed SSB degen DH10b failed to grow - Either Gibson failed (probable from ss => ds conversion vs amplify results) or transformation failed - Retransform with +ve control (pBAD) Will Gibson once with amplified dsDNA product and then proceed with selection on Tues
19-8-26	KA	Transformed DH10 beta with Gibson product, Gibson Pos Control, Empty pBAD, water control. Did not flick the tube after plasmid has been added Incubated gibbon product, empty pBAD & water control at 37

		celsius in 5 ml LB broth + 5 microL tet, awaiting results; gib pos was incubated in ampicillin Plated 1:10, 1:100, 1:1000, 1:10⁴, 1:10⁵ dilutions of Gibson Product, 1:1 empty pBAD, 1:1 water control on tet plates, yesterday's 1:1 Gibson positive control on amp plates, today's 1:1 Gibson pos control on amp plates. Received 0.22 microM Millipore filter paper from Sio for desiccation troubleshoot - will try this out tmr Worked on presentation and finalised parts
19-8-26	YF	Loaded: - 2nd linear DPN1 digest extracted product: failed - wrong voltage - epPCR R2 gel extracted product: faint - epPCR R3 purified vs unpurified troubleshoot: purified is better, unpurified produces 100 bp product - epPCR R3 sample 9 & 10 - epPCR R3 sample 11, 12, 13 & 14 - ss => ds conversion vs amplified product: amplified better, clear band, conversion failed (smear) Made 1 x 100 mL 2% Li3BO3 prestain gels (2g, 100 mL, 5 uL SYBER) Made 6 x 35 mL 2% Li3BO3 prestain gels (5g, 250 mL, 12.5 uL SYBER) Gel Extracted Sample 9 & 10
19-8-26	CW	Conduct ss to ds amplification instead of simple ss to ds one cycle conversion. Decided to run amplification after the ds conversion. This increase the dsDNA degenerate library concentration for easier gel extraction and gibson assembly. Continued with epPCR proceed to round 3. Decided to run multiple replicates and use different gel extracted template tubes to increase the chance that at least one tube contains optimal amount of correct amplicon and no/little by-product and smears. The correct original template for NEXT IDP arrived so epPCR started with NEXT as well. NEXT showed comparatively less smearing than SSB during epPCR.
20-8-26	General	Gibson with ss to ds amplified conducted - Realized Gibson reactions do not include extra master mix for positive control - Plated to see if succeeded tmr epPCR SSB R5 & NEXT R2 conducted
20-8-26	KA	Loaded gels for epPCR products (SSB R4 and NEXT R2)

		Finalised protocol for heat shock Finalised protocol for osmotic stress Since the prev filter paper curled up after ethanol has evaporated, gonna use 0.66 microM filter papers instead.
20-8-26	YF	Performed Gibson assembly on SSB degen amplified product Extracted Sample 2 & 3N from SSB R4 and NEXT R1 epPCR
20-8-26	CW	Conducted epPCR for SSB up to Round 5 (this is the highest round we have achieved). Conducted epPCR for NEXT to round 2. But the product already smeared very severely and brightness of the band on gel is faint. This lead us to think that perhaps using this unusually short template inherently is accompanied with replication slippage, increase in the longer size by-product, smearing, and reduction of amplifiable template over rounds. Helped with instructions and calculations in Gibson assembly of SSB degenerate sequence into pBAD+betalac.
20-8-26	James	Transformed Gibson assembly products into DH10b competent cells. Cells from previously failed gibbon reaction (assembly of SSB library) were reincubated and plated in LB over the long weekend.

Date	Person	Log
25/8/26	General	Gibson 21/8 Plate results: - Gibson +ve control had no colonies - Gibson liquid culture had growth - suspected contamination by pBAD/beta-lac insert - reculture in 300 ug/mL Amp - Reculture grew => Gibson +ve control exists but unknown if contaminated - Colony count (or lack thereof) doesn't make sense: suspected murky is due to contam, but its not - still unresolved - Water -ve control had many small colonies - SSB library has many small colonies - pBAD +ve control has small and large colonies Contamination suspected; moving forward: - LB flasks will be labelled with date used and incubated overnight after use to detect contamination - Workbench & pipettes will be sterilized daily - Pipette tips have been swapped for known sterile tips Gibson redone by Alanna in her lab - 2nd linear digest PCR purified by her, 43 uL purified - If Gibson successful, trial will be conducted on degen library, thus selection round occurs in W9 - Colonies seen next day. Gel kept over 3 days extracted (SSB R5 & NEXT R3) - Didn't show up on gel => prob degraded - Appears on Nanodrop: products used for downstream processes to see if viable Target Protein is being finalized
25/8/26	James	Gibson negative ctrl plates showed small colonies. pBAD control showed large and small colonies. Gibson position control plates showed no colonies, but liquid culture contained cells. These cells were reincubated in 300ug/ml Amp and grew (OD increased 0.982 to 1.384 in 2 hrs). Plasmids were extracted. SSB plates (gibson products) showed no colonies on plates but cells in liquid culture. However since negative control failed we have no way of telling if these are successful cells Alanna will help with gibson assembly from this point. We will abandon troubleshooting for this stage and aim to send a constructed library for sequencing by Friday
25/8/26	YF	Made 1 x 50 mL TAE prestain gel Made 5 x 35 mL Li3BO3 prestain gel Loaded:

		- epPCR SSB R5 & NEXT R2 gel extracted product
25/8/26	CW	epPCR continued, nanodrop showed gel extracted SSB R5 and NEXT R3 seems to contain DNA. However, no product was able to be further amplified from SSB. NEXT R3 sample underwent simple PCR to increase template concentration, and was able to undergo further epPCR.
25/8/26	SS	Research about target protein= TEV protease a possibility as used in many papers, shown to be improved in solubility, has interesting applications in engineered cell signalling, small monomeric and shown to be improved in solubility and can be used to also determine how our tags affects enzymatic activity as it is an enzyme.
25/8/26	KA	Conducted gel extraction on SSB R5 and NEXT R3 Nanodrops results were promising For SSB R5, 6 nanodrop results were acquired; eventually settled on 3 concordant results: 5.1, 6.0, 5.7 ng/microL; 260/280 was within the range of what we normally get → tube labelled "6" Tube labelled "1" and "3" → Tomn did the nanodrop, with one of the tubes being significantly more concentrated (30+ instead of the usual 5 to 8) → suspect added the wrong elution buffer as the surface tension on the nanodrop seemed off, hence, redid the whole process on tube "6".
26/8/26	KA	Conducted purification using NEB's kit on the amplified ssDNA→dsDNA products - nanodrop result: average 12.9 nanogram/microL with 260/280=1.8 and 160/230=2.0 Read papers related to the tardigrade system for the final target protein → decided to do citrate synthase & LDH based on literature.
26/8/26	SS	- Poured 20 Lennox tetracycline plates - Continued research about target protein , TEV protease likely difficult to use as it is classed as a plant virus protein, however shortlisted two other possible candidates IL3 and MMP13, tense cannot be used for enzymatic activity assay however have been shown to improve in solubility in other papers, MMP13 particularly have only been shown to improve in solubility by two tags Nus A and SUMO and not affected by some other conventional tags such as His6 and MBP. - Passed information onto Tomn to make final decision
26/8/26	James	- Reincubated 3 strains: Dh10b with pBAD, DH10b with pBAD+blac insert, BL21 with pBAD. Will make glycerol stocks tomorrow - Performed gibson assembly of SSB library using Kean Anns purified library

		<table border="1"> <tr> <td>backbone</td><td>2.1ul</td><td>0.033 pmol</td></tr> <tr> <td>SSB lib</td><td>1.3ul</td><td>0.167 pmol</td></tr> <tr> <td>ddH2O</td><td>6.5ul</td><td>-</td></tr> <tr> <td>Master mix</td><td>10.0ul</td><td>-</td></tr> <tr> <td>total</td><td>20.0ul</td><td></td></tr> </table> According to NEB calculator. Positive control: 2.5ul master mix + 2.5ul gibson positive ctrl Accidentally added tet instead of amp into positive control. Not expecting to see growth in liquid culture Update: plates had very little colonies, but liquid culture of pBAD and SSB library showed cloudy. Will extract plasmids for sequencing	backbone	2.1ul	0.033 pmol	SSB lib	1.3ul	0.167 pmol	ddH2O	6.5ul	-	Master mix	10.0ul	-	total	20.0ul	
backbone	2.1ul	0.033 pmol															
SSB lib	1.3ul	0.167 pmol															
ddH2O	6.5ul	-															
Master mix	10.0ul	-															
total	20.0ul																
26/8/26	YF	Loaded: - pBAD/beta-lac backbone: Gel vs PCR purified samples - epPCR NEXT R3 and SSB R5, R3 PCR Gel Extracted: - epPCR R3 NEXT (1 & 2) - SSB R3 PCR Researched target proteins for validation															
27/8/26	KA	Extracted plasmids for SSB Library and SSB Original. SSB Library was Gibson-assembled and transformed by James while SSB Original was Gibson-assembled and transformed by Alanna. 15 microL SSB Library was sent off for sequencing, awaiting results. Nanodrop results are promising: 47.0, 44.1 and 41.7 ng/microL for SSB Library and 46.9, 78.6, 49.7, and 48.6 ng/microL for SSB Original. 78.6 is likely due to not flicking the tube.															
27/8/26	SS	DNA purification with our original kit. 90 uL of each of the SSB, De Novo and NEXT tags were purified, while 30 uL of 2nd linearisation pBAD DpnI digested product was purified.															
27/8/26	YF	Plasmid Extracted: - pBAD - pBAD/beta-lac															
27/8/26	CW	Continued mainly with epPCR works Picked new colonies of DH10 beta cells from the initial plate we were growing and incubated them into falcon tubes overnight. This is to make fresh new DH10 beta competent cells.															

28/8/26	YF	Made 3 x 0.75% TAE prestain gels Made 3 x 2% TAE prestain gels Loaded DPN1 digest 2nd linear - failed
28/8/26	CW	Nanodropped "KA 26/08/2026 Monarch NEB purified" SSB dsDNA library, diluted to the correct concentration, and send the ss-to-dsDNA amplified SSB degenerate library for amplicon library sequencing. This was to see whether the conversion process introduced too many extra nucleotide errors because our starting template was ssDNA library not a single ssDNA species. This determined that further gibson assembly on the converted dsDNA SSB library is viable.
28/8/26	SS	Nanodrop results Average concentration for degenerate De Novo, SSB and NEXT tags after eppcr were 31,38 and 31 ng/uL respectively. Average concentration for purified second linearisation of DpnI digest was around 25 ng/uL Nanodropped gel extract products for SSB and NEXT, concentrations using NEB kit very low around 2 ng/uL, collective decision to use another kit. Nanodrop pBAD and pBAD plus Blac Calculated volume of each tag and backbone needed for Gibson assembly for both degenerate tags and original NEXT and SSB tags.
28/8/26	KA	Made DH10 beta competent cells from Tomn's overnight cultures Centrifuged the cells for 15 minutes in the cold room for getting rid of the LB and washing the cells with CaCl₂ Only the first two tubes were used as the 3rd one contained some mysterious black substances floating around Tube 2 was grown to 0.527 (faster growth because of the smaller initial starting volume? Because the final volume was only 41ml, while tube 1 had 45ml) Tube 1 was grown to 0.455 Incubation time: around 2.5 hours (10:50am to 1:25pm) They were placed in ice at 1.25pm for 40 minutes instead of the "20 minutes" written on the protocol. Control stayed consistently at around 0.060, similar to the blank (pure Miller LB) Miller LB was used throughout because of insufficient Lennox LB available As the autoclave machine broke down ("minimal pressure error") the solution containing CaCl₂ was not autoclaved. Instead a 0.22 microM filter was used to filter it.
28/8/26	James	Checked competence of our competent cells stocks. Batches of DH10b and B121 strains were transformed with pBAD according to the protocol below: - Add 0.5ul of plasmid DNA to thawed cells - Leave on ice for 30 mins - Heatshock @ 42C for 45s

		- Leave on ice for 2 mins - Take 25ul aliquot of cells, add 75ul LB, plate immediately - Take remaining 25ul of cells, add 975ul of LB, incubate for 1.5hrs - Plate 100ul of incubated cells Results show that transformation efficiency was higher for DH10b than for BL21, though BL21 tends to form more massive colonies Ideas from advisors to improve transformation efficiency: - Less time left on ice after addition of plasmids, maybe 5-10 mins - Add more plasmids - Electrocompetency (not possible)
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Date	Person	Log
1/9/26	General	Gibson inserts measured - Insufficient backbone to proceed - Linearization, digest, purification and validation redone - Significant degradation of inserts - ss to ds amp, purification and validation redone epPCR NEXT R5 & SSB R4 failed => suspect due to master mix freezing => precipitating Mn 2+ Plate results from 28/8 on competent cells obtained - All cells competent - BL21 has very low colony count => prep to make new batch of competent BL21 Discussion on validation round oligo ordering - Debating plasmid vs G-block ordering - Plasmid: no gibson req; limited repertoire of target protein - G-block: req gibson; full repertoire of target protein available
1/9/26	KA	Inoculated 3 colonies of WT untransformed BL21 from Rachel's plate from 22/7/26 + 1 control to prepare an overnight culture for making BL21 competent cells as the colony count is low from James' plates from 28/8 → around 5 to 8 fat colonies, with 1 plate having no colony at all Nanodropped SSB's R3 gel extracted products, NEXT R4 gel extracted products, pBAD insert IDP (F), pBAD insert IDP ® and pBAD beta-lac backbone. Conducted purification on the PCR amplified NEXT, De Novo and SSB degenerate sequences.
1/9/26	YF	Loaded: - epPCR NEXT R5 & SSB R4: Smear => failed - Purified vs unpurified degen amp: validated amplification & purification, tho mix up suspected - Linearized Backbone: band seen => validated linearization, digestion and purification Made x20 Lennox LB Broth (50 mL) Made x1 TAE 2L Nanodropped Gibson inserts => concluded amounts insufficient to proceed Made x1 Li3BO3 2L

1/9/26	SS	Purified three samples Used a total of 277 ul of each sample for DNA purification The samples were 1. DpnI digested first linearisation product 2. DpnI digested second linearisation product from old sample (forms source 17/8) 3. DpnI digested second linearisation product form new sample today All samples were digested using new NEB DpnI digestion kit Results from 2/9 = concentrations of backbone were above 100 ng/uL because of the high volume used, and the 280/260 and 260/230 values were more optimal.
1/9/26	James	Performed DpN1 digests on previously obtained linear PCR products: - 17/7 first linearisation - 30/7 first linearisation - 17/8 second linearisation Using NEB DpN1 kit since we ran out of the thermofisher one. Sent digests to be purified by Setareh.
1/9/26	CW	Explored validation round target protein-IDP fusion protein plasmid construction options. Make preliminary oligo orders for plasmid construction. Identified and explored the option of directly ordering circular plasmid without gibson assembly. That option is not very viable because the estimated arrival time is very late and we can only use very short genes of target protein (probably only heGF-GB1 fusion protein etc.) Also planned to do validation only on the original sequence of IDPs because by the time we get the selected sequence, ordering the oligos would not arrive in time. epPCR redo on round 5 for NEXT and round 4 for SSB (to start with a less smeared template and also previous round 5 seems not recoverable by PCR—probably amplicon damaged or degraded by repeated freezing?)-->both did not work. Suspect that the master mix was freezed too long and might have the Mg²⁺, Mn²⁺ ions precipitated or Taq activity decreased. Did ss to dsDNA conversion + amplification again. The ssDNA are not fresh out from the manufacturer's tube but from diluted tube so there might be mixing up of different IDP. All using high-fidelity Q5-XT hot start enzyme. See a small degree of smearing. Probably smearing on the gel for such small amplicon is inevitable for epPCR.
2/9/26	General	Selection round begun: - Gibson products produced and transformed

		- Transformed cells seeded into 50 mL instead of 15 mL (will correct protocol) - Currently, seeded flasks have not had significant growth, will continue tmr New DH10b competent stock transformed to gauge competency epPCR continues to fail - Template does not appear => PCR amplify template & revert to previous round Results of ss => ds amp from different days shows mix up between vials (suspected mislabelling in diluted stock)
2/9/26	YF	Nanodropped Gibson inserts and performed gibbon Loaded: - epPCR NEXT R5 SSB R4: still fails - not master mix - Unpurified ss -> ds amp (1/9 & 27/8) => mix up confirmed - Purified ss -> ds amp & epPCR NEXT R4 PCR SSB R3 Gel extraction product: confirmed mix up is not loading error; template for NEXT R5 & SSB R4 in question Researched functional assays for mTdT, taCA, isPETase
2/9/26	CW	epPCR NEXT round 5 and SSB round 4 failed again. Ss to ds DNA amplification of all IDP library was confirmed to contain swapping of IDP identities. Tried to amplify NEXT round 3 and SSB eppcr product by Q5 to recover it.
2/9/26	KA	Used FIJI to count colonies on 0 hour DH10beta competency check - Stock 1: 310 colonies - Stock 2: 665 colonies - Stock 3: 751 colonies Researched on tags that our IDP tags can be benchmarked against: SUMO, MBP, NusA and Glutathione S transferase Picked SUMO because it is a small protein and MBP is larger MBP: used in the NEXT paper for benchmarking SUMO: small protein that makes expressed proteins soluble in E coli, is ubiquitin-like. https://pmc.ncbi.nlm.nih.gov/articles/PMC2242369/ https://link.springer.com/article/10.1023/B:JSFG.0000029237.70316.52 Setareh found a paper: https://www.sciencedirect.com/science/article/pii/S1046592810001373 that says that Human SUMO isn't well characterised

		and its use as a tag hasn't been researched on Paper that Tomn sent uses SMT3; sequence: 77 <table><tr><th colspan="4">Table 1. Cloning details for the constructs used in this work.</th></tr><tr><th>Target vector/ digested with</th><th>Cloned gene</th><th>PCR primers and restriction endonucleases used to digest the product</th><th>Resultant plasmid</th></tr><tr><td>pET24d Nco I Bam HI</td><td>6xHis-SUMO (Smt3)</td><td>F tttttccatgggtcaccatcatcatcacgggtcgactcagaagtcaatcaa R ttttttgatccgggtctcaacctccaatctgttcgggtgag Nco I Bam HI</td><td>pET24-6xHis-SUMO</td></tr><tr><td>pET24d Nco I Bam HI</td><td>6xHis-Ub</td><td>F tttttccatgggtcaccatcatcatcacgggcagatctttgtgaagaccctc R ttttttgatccgggtctcaacctctcgcaggcgcaacaccag Nco I Bam HI</td><td>pET24-6xHis-Ub</td></tr></table> NusA: Large & compatible but we want variety so SUMO is picked instead. GST: Mammalian (used to make drugs soluble so can be easily flushed out by the liver) so might not be compatible with the e.coli system. + will also take part in nucleophilic reactions in cells as its native state combats oxidative stress in mammalian cells Won't need to cleave the proteins off in the end GNN as a backup for computational selection? Will need to research this further as the paper Tomn found: https://www.sciencedirect.com/science/article/pii/S2666386424006064?via%3Dihub is for liquid-liquid phase separation, where aggregation is induced by hydrophilic interactions (thru hydrogen bonds) and not hydrophobic interactions (which is what our aggregation system is based on)	Table 1. Cloning details for the constructs used in this work.				Target vector/ digested with	Cloned gene	PCR primers and restriction endonucleases used to digest the product	Resultant plasmid	pET24d Nco I Bam HI	6xHis-SUMO (Smt3)	F tttttccatgggtcaccatcatcatcacgggtcgactcagaagtcaatcaa R ttttttgatccgggtctcaacctccaatctgttcgggtgag Nco I Bam HI	pET24-6xHis-SUMO	pET24d Nco I Bam HI	6xHis-Ub	F tttttccatgggtcaccatcatcatcacgggcagatctttgtgaagaccctc R ttttttgatccgggtctcaacctctcgcaggcgcaacaccag Nco I Bam HI	pET24-6xHis-Ub
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2/9/26	James	Performed transformation of the following into DH10b cells: - NEXT tag (original) - NEXT tag (library) - De Novo (library) - SSB (original) - SSB (library) - pBAD (into existing stock) - pBAD + blac - pBAD (into new stock 1) - pBAD (into new stock 2) Modifications from previous transformation protocols: - 15 mins ice after adding DNA instead of 30 - 1ul DNA added instead of 0.5 - Incubate in 1ml LB after heatshock, then incubated into 50ml conical flask of LB - OD was monitored at 90 min intervals. Originally planned to grow until 0.3 OD then spin down to prevent further growth skewing library. However OD reached only about 0.05 after 3 hrs. We decided to store the libraries in the fridge overnight and restart incubation tmr																
3/9/26	General	Gibson Plates Results: - Gibson and transformation successful - NEXT degen has little to no colonies (suspect just very small population diluted too much)																

		50 mL seeded flask haven't grown in 5.5 hours in incubator - accept risk of skew and incubated overnight epPCR continues to fail - Decided to stop and move forward with current highest round product with cut off from R3 BL21 were made competent
3/9/26	James	Performed Gibson assembly & transformation for Next library using ssDNA oligos (bypass the step of creating dsDNA and amplification) Incubated a colony from De novo degen for sequencing tomorrow to confirm that gibbon has occurred. If overnight cultures of all libraries fail, we will retransform tomorrow and incubate over the weekend. If that fails, we will redo the gibbon next monday.
3/9/26	CW	Continued with eppcr but cannot recover above round 4. Likely the product in round 4 or 5 already has too much by-product and was not able to recover any useful correct amplicon. Round 3 was determined as the certain experimental upper limit of doing epPCR and sent off for sequencing to calibrate the amount of mutations in the template. Only NEXT epPCR product underwent library sequencing due to budget constraint.
3/9/26	YF	Conducted OD measurement of seeded selection round samples - No growth after 5.5 hrs, leave in incubator overnight Plated BL21 competent stock for sampling Made 5 x 35 mL 2% Li3BO3 prestain gels Loaded - epPCR R2 SSB (smear); PCR NEXT R3 - gel extract (faint - Qiagen) - PCR SSB R2 (smear) - PCR SSB R4 (smear); PCR NEXT R3 (succeeded)
3/9/26	SS	Gel extraction on two NEXT PCR round 3 samples N1 and N2 Results around 8.0 ng/uL for both samples when nanodropped.
3/9/26	KA	Made more BL21 competent cells Control incubated overnight, the next day: cloudy But the control stayed consistent over the OD measurements i.e. around 0.001 to 0.008 while the rest reached 0.4 to 0.5. Might be contamination when pipetting things

4/9/26	KA	Plated 9 plates to check transformation efficiency of BI21 Gel extracted De Novo library Monoclonal + SSB library that has been incubated overnight → sent SSB lib for sequencing; nanodrop results: 161 ish ng/microL Asked sio about L-LDH → she has the Type II one, suspended in Ammonium sulfate. However the amount is small and she might not be able to give it to us as she needs it for the practicals; if she places the order soon it'll only arrive on Sept 18th.
4/9/26	James	Performed selection. Each tag library was seeded into 50ml of media containing carbenicillin concentrations ranging from 0 to 3000 ug/ml, as well as 0.2% arabinose and 10ug/ml tetracycline. OD600 measurements were taken at 0hrs, 30mins, 1hr, 1hr45. Control flask containing only stock pBAD plasmids showed decrease in density, while all other flasks including the untagged b-lactamase showed increase in density. We concluded that the concentration of antibiotic is not enough to suppress bacteria growth likely due to using AraC promoter to express b-lactamase. Possible workarounds: - Increase carbenicillin concentration - Decrease arabinose concentration Will retry selection under new conditions next week
4/9/26	YF	Incubated strains in 50 mL of LB - did destructive and non-destructive OD600 readings of strains on 96 well plates - found that non-destructive sampling yielded no significant differences - proceed with destructive sampling for all future readings Incubated strains to 0.3 OD600 and induced to 0.6 Performed carbenicillin selection - pBAD control died in presence of carbenicillin => carbenicillin is exerting a selection pressure - pBAD/beta-lactamase control did not have significant difference within concentrations => - insufficient time to act on the semi-resistant cells => run selection for longer - beta-lactamase has better performance than expected due to either mutation or false reporting => no way to fix (unlikely as sequence has been validated to be correct) - beta lactamase expression is not the same and thus MIC is higher than expected => most likely explanation as we used different promoter systems - several assumptions were proven: - dead cells do not contribute to OD and cell

		fragments have minimal contributions (OD can decrease) - significant decrease in initial population is generated to produce significant differences in final population
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Date	Person	Log
7/9/26	James	Performed selection again, adjusting carbenicillin concentrations to be from 5000 to 40,000 ug/ml. OD600 readings were taken at the same time intervals. Results showed decrease in density. Cultures at 40,000, 30,000 and 15,000 were streaked at 1hr 45mins, showing large numbers of colonies
7/9/26	Setareh	PCR purification of tardigrade degenerate sample after single strand to double strand conversion
7/9/26	KA	Revised protocols for tardigrade selection
7/9/26	YF	Repeated selection with lower seeding density and higher concentrations - estimated hat the max MIC of WT beta-lactamase was around 32k ug/mL - liquid batch used only 2 mL of stock to minimize carbenicillin usage
7/9/26	CW	Ss to dsDNA conversion of degenerate library of tardigrade fragment. Design gibson assembly calculations, protocols, etc.
8/9/26	Setareh	Made 40 agar plates with tetracycline and Lennox LB
8/9/26	KA	Did Gibson Assembly on tardigrade original, tardigrade degenerate and Positive Control and plated them. Ran out of time to seed 50 ml LB broth, had to retransform tmr. Plate results: Gibson assembly of dsdna failed. Had to troubleshoot.
8/9/26	YF	Selection continues from 7/9 - no change in OD - samples streaked on carb. plates Based on OD data, samples were taken to incubate for extraction: - DN: 0, 10k, 15k, 30k - SSB: 0, 30k - NEXT: 0, 15k Attempted plasmid extraction - failed based on OD data Conducted LB vs Saline OD600 measurement experiment - used to benchmark effect of saline in taking OD600 measurements - no significant difference was found
9/9/26	Setareh	Plasmid extraction of 8 samples of tags with varying concentration of ampicillin DN 0,10,15,30 NEXt 0 and 15, SSB

		0,30 PCR purification of of round 4 NEXt 1+2 and round 3 SSB 1+2, results on nanodrop very inconsistent but might be around 10 ng/UI for both, however the problem might be with nanodrop because elution buffer concentration also varied from -7 to 2 with multiple measurements.
9/9/26	YF	Made 5 x 35 mL 2% prestain Li3BO3 gels Made 14 x 50 mL Lennox flasks Continued incubating selected samples from 8/9 across the day Nanodropped components for tardigrade gibson assembly
9/9/26	James	Performed 3rd batch of carbenicillin selection with the same conditions used as 7/9/26
9/9/26	KA	Did tardigrade dry run with isopropanol+dry ice. Plates show a nice decrease. But now there is a concern that isopropanol could have gone in and affected results. Solution: wrap it in Parafilm and re-try soon. Placed freezer-dry-run cells in the freezer at around 4 ish. Plated at 3 ish on 10th of sept. Awaiting results on 11th of Sept. T0 plated alongside flash-freeze dry run.
10/9/26	Setareh	PCR purification of ss to dsDNA conversion of tardigrade, resulting nanodrop is around 14-16 ng/uL Plasmid extraction of original tardigrade DH10B , concentration when nanodropped between 35-40 ng/uL Plated freezer dry run sample 1;1, 1:10, 1:100 and 1:1000 dilution.
10/9/26	YF	General organization of lab space done Sequencing results showed skewing in 0 conc batch - redo Gibson assembly
10/9/26	KA	Gibson Assembly was re-done by Tomn with a few changes 1. 1 hour incubation time 2. PCR purified dsDNA degenerate 3. ssDNA degenerate sequences Transformed and plated 5 things: 1. dsDNA degen 2. ssDNA degen 3. Water negative control 4. Gibson Assembly Positive control 5. Empty pBAD Incubated 5 flasks 1. dsDNA

		2. ssDNA 3. "Nothing" but with tet to check for contamination 4. pBAD 5. Original (inoculated from James' plate from yesterday)
10/9/26	James	Sequencing results arrived - What we thought was SSB library is actually NEXT library, and is also monoclonal - What we thought was NEXT library is actually a plasmid with no insert - De novo library sequencing failed. No usable data was generated Actions forward: - Reincubate all cultures at different carbenicillin concentrations, miniprep tomorrow - Redo selection next monday (start from gibson assembly). Rethink about how we are going to approach the problem of skewing to avoid getting monoclonal cultures
10/9/26	CW	Gibson Assembly was re-done by Tomn with a few changes 1. 1 hour incubation time 2. PCR purified dsDNA degenerate 3. ssDNA degenerate sequences 4. More methodologies can be found in report and supplementary materials.

Date	Person	Log																
14/09	Setareh	Converted gel image files from SGD to TIFF and from TIFF to PNG using a script as only PNG images can be uploaded to google doc for report. Uploaded some good images into the results section of the report. Adjusted contrast and settings of the image so the gel image is more clear and bright.																
14/9	KA	OD600 measurements were taken of the cultures that have been stored in the fridge over the weekend using the 96-well plate reader. Library's OD: 0.084, 0.089, 0.092 With 1:10 dilution, Tardigrade CAHS Core: 0.064, 0.065, 0.066 Empty pBAD: 0.083, 0.083, 0.083 The blanks have been subtracted from these values. The cultures will then be diluted to 0.02 (with the blanks subtracted) so that the cells can start growing at similar OD600 and be "woken up". The total volume of the culture will be 10ml. <table><tr><th>Variant</th><th>Old culture volume/ ml</th><th>Fresh media/ ml</th><th>Final volume/ ml</th></tr><tr><td>Library</td><td>4.65</td><td>5.35</td><td>10</td></tr><tr><td>WT</td><td>0.87</td><td>9.13</td><td>10</td></tr><tr><td>pBAD</td><td>0.48</td><td>9.52</td><td>10</td></tr></table> OD600 measurements were taken every 30 minutes to 1 hour. The WT and pBAD were induced at 2:05 pm and kept on ice. At 2:50 pm, ssDNA had an OD of 0.077, 0.076, 0.079; while WT's OD was 0.179, 0.181 and 0.184 and pBAD's OD was 0.179, 0.179 and 0.189. ssDNA was induced as well with 1% l-arabinose later in the day. However, the experiment was not successfully carried out today and the stocks were kept in the fridge.	Variant	Old culture volume/ ml	Fresh media/ ml	Final volume/ ml	Library	4.65	5.35	10	WT	0.87	9.13	10	pBAD	0.48	9.52	10
Variant	Old culture volume/ ml	Fresh media/ ml	Final volume/ ml															
Library	4.65	5.35	10															
WT	0.87	9.13	10															
pBAD	0.48	9.52	10															
14/9/26	YF	Conducted gibson assembly for all ssDNA libraries (EB tags) Gibson products transformed and seeded into 50 mL LB and incubated at 37 celsius for 8 hours Conducted equipment standardization experiment - measured x2 serial dilutions on both 96 well plate reader and 1 cm pathlength spectrophotometer - found multiplicative correction factor x5.5 for sequence																

		Continued research on solubility assay
15/9	KA	The OD's of the stock solutions from the fridge were measured again. The OD's of pBAD and WT were assumed to be the same as the first time the OD's of their stock solutions were measured. The OD's of the Library induced culture left in the fridge overnight were measured. There was not enough stock solution to use fresh, uninduced Library. However, the cells were washed with LB and re-incubated before selection to remove all traces of L-arabinose and its effect. Everything was diluted down to 0.02 (with the blank subtracted) on the 96-well plate reader. The first OD measurement was done at 12:47 pm, and these are the OD measurements for different variants: pBAD: 0.062, 0.067, 0.064 WT: 0.068, 0.064, 0.067 Library: 0.061, 0.065, 0.071 It was then incubated at 12:50 pm, and OD measurements were taken roughly every 1 hour. Glycerol stocks of the library were made as well. Before induction, the OD measurements are: Library: 0.085, 0.086, 0.085 pBAD: 0.155, 0.159, 0.158 WT: 0.162, 0.172, 0.163 pBAD and WT were diluted to half of their initial OD, i.e. 5ml of fresh LB was added to 5ml of the culture. The cultures were then induced at 4:25 pm at 25 °C The amount of L-arabinose added: Culture volume of ssDNA: 7.5 ml, 395 µl of 20 % L-arabinose was added Culture volume of pBAD and WT: 10ml, and so 526 µl of 20% L-arabinose was added. The cultures now contained 1% L-arabinose. They were then incubated at 25 °C overnight.
15/9/26	YF	Continue incubation of seeded colonies Prepared carb tubes for selection Made 1 x 50 mL 0.75% prestain TAE gel
15/09	Setareh	Made 20 tetracycline plates Began working on COSHH forms using SDS documents
16/9	KA	The OD measurement was taken Library: 0.146, 0.172 WT: 0.163, 0.153

		pBAD: 0.170, 0.165 The selection started at 12:20pm. The cultures were washed with saline and then suspended in saline. They were then divided into aliquots of 1.0ml aliquots and subjected to freeze-thaw in the isopropanol/dry ice mixture. They were frozen for 15 minutes, and thawed for 25 minutes. They were inspected visually between cycles to ensure complete thawing and freezing. They were also wrapped in a layer of parafilm to ensure the eppendorf cap did not pop open during the experimental process. 6 cycles of freeze-thaw were completed that day. The cells, after each cycle, were serially diluted and plated.
16/09	CW	Attempted to amplify out the IDP sequences from the library of plasmids using IDP F and IDP R primers (then later the pBAD F and pBAD R primers) This would allow the sample to be send for PCR amplicon library sequencing so we can see the individual variants retained. Different PCR conditions were run to find the optimal condition that (1) does not introduce errors in replicating (2) proportionally increase all variants in the post-selected library [no skewing] (3) have sufficient yield for gel extraction (because there were clearly unremovable mispriming products). Tested primer concentration, GC enhancer concentration, template concentration, annealing temperature, etc.
16/09	Setareh	Gel purification of three samples SSB fished out tag and SSB and NEXT epPCR, with the eppcr samples having around 8-10 ng/uL of concentration and the SSB fished out tag having a concentration of around 13 ng/uL. Completed COSHH forms for all current and future reagents.
16/9	James	Streaked CAHS freeze-thaw plates (cycles 0-6) on arab plates. Various degree of glowing (pBAD contamination): less in lower cycles, more in higher cycles
16/9	YF	Loaded 1st linear purified digest backbone - succeeded Loaded epPCR R2 NEXT, SSB, SSB fished Conducted selection - lower than intended seeding density - no actionable OD600 data Induced tardigrade strains Seeded 4.5 mL LB with 500 uL of tag stocks (0, 30k, 40k) - 1 set were incubated at 37 Celsius - 1 set were incubated at 25 Celsius

17/9	YF	Checked OD readings of 16/9 incubation - no growth for 30k and 40k across both sets Conducted another round of selection with correct seeding density - 500 uL of each 0, 15k, 30k, 40k incubated in 4.5 mL LB
17/9	James	Conducted another round of carbenicillin selection with correct seeding density
17/9	KA	A second round of flash-freeze was carried out with the new batch of Gibson Assembly. After induction, the OD600 (via 96 well plate) of the tubes before selection were: WT: 0.148, 0.143, 0.130 pBAD: 0.158, 0.141, 0.126 Lib: 0.132, 0.114, 0.134 Cryo tubes were used instead of eppendorf tubes. The selection protocol was the same as yesterday's. However, this round of selection was rendered invalid because of the following reasons: 1. Some tubes were mixed up with the incorrectly done ones. 2. 3rd cycle onwards, the tubes were not completely thawed in the 2nd cycle before they were frozen, and the subsequent tubes were thawed for 35 minutes to ensure complete thawing. The cells were then serially diluted and plated. 500 µl of aliquots from the same culture were also made for deep-freeze selection, they were then placed in the freezer. The tubes were frozen for some time before they were thawed for 45 mins to 1 hour at room temperature. They were then serially diluted and plated.
18/9	YF	Made buffers for SDS Page gels - Gel Buffer (x3) - Cathode Buffer - Anode Buffer Made 5 x 12% Tris Tricine gels
18/9	KA	Imaged and counted the colonies from the selection rounds Plated deep-freeze cycle 2 and 3 Replated selected pBAD and WT from Gibson Assembly Batch 1 that have been sitting on the bench for 2 days at room temperature suspended in saline. Also made 3 x 12% Tris Tricine gels
18/9	CW	Nanodrop, start preparing some samples of DNA library for sequencing.

Date	Person	Log
21/9/26	YF	Made lysis buffer for chemical lysis Loaded Gel & PCR purified epPCR R3 samples Conducted chemical lysis
21/9/26	KA	Plated deep freeze cycles 3 and 4 Did plasmid extraction on the entropic bristle tags (0k and 40k)
21/9/26	CW	Helped with using a transilluminator to image E coli plates with pBAD (sfGFP containing) plasmid contamination.
22/9/26	James	Performed plasmid extraction and purification on samples to be sent for sequencing. Gel electrophoresis shows faint bands indicating mispriming/impurities. Decided to gel extract band at correct position to ensure accurate sequencing
22/9/26	CW	Helped with gel electrophoresis, imaging of gel, and do more amplifying out of IDP libraries that need to be send for sequencing.
22/9/26	YF	Conducted mechanical lysis via freeze thaw and mechanical bead disruption Nanodropped supernatant concentration for all fusions Ran SDS-Page gel for x1, x10, x20 dilutions of GB1-isPETase samples Fixed, Stained, Destained gels before imaging
23/9/26	James	Performed gel extraction on PCR products for sequencing.
23/9/26	YF	Ran SDS-Page gel for x10 dilution of all fusions with washings of insoluble pellet Buffers for protein purification were prepared
23/9/26	KA	Performed plasmid extraction on the entropic bristle tags (0k, 15k, 40k), 12 tubes in total
23/9/26	Setareh	Plasmid extraction for tardigrade cycle 6 to send off for sequencing Look through the report discussion and edit
23/9/26	CW	Amplify out even more IDP libraries to be send off for sequencing, load the PCR product into and ran gel electrophoresis (2% lithium borate gel, 120V for 90 minutes), cut out the correct amplified product for gel extraction.
24/9/26	James	Performed gel extraction on PCR products for sequencing.

24/9/26	YF	Buffers for protein His-Tag Spin Trap purification was prepared Proteins were purified All washings were loaded, ran, fixed and stained
24/9/26	KA	Nanodropped samples that would be sent for sequencing.
24/9/26	CW	Amplified out more IDP libraries ready for sequencing, helped with nanodrop and adjust DNA concentration for sequencing samples. Final IDP library fished out was tardigrade freeze-thaw cycle 6, loaded its gel and cut out the gel and extracted. This last sample was off for sequencing on 28/9/26
25/9/26	YF	Gels were destained then imaged